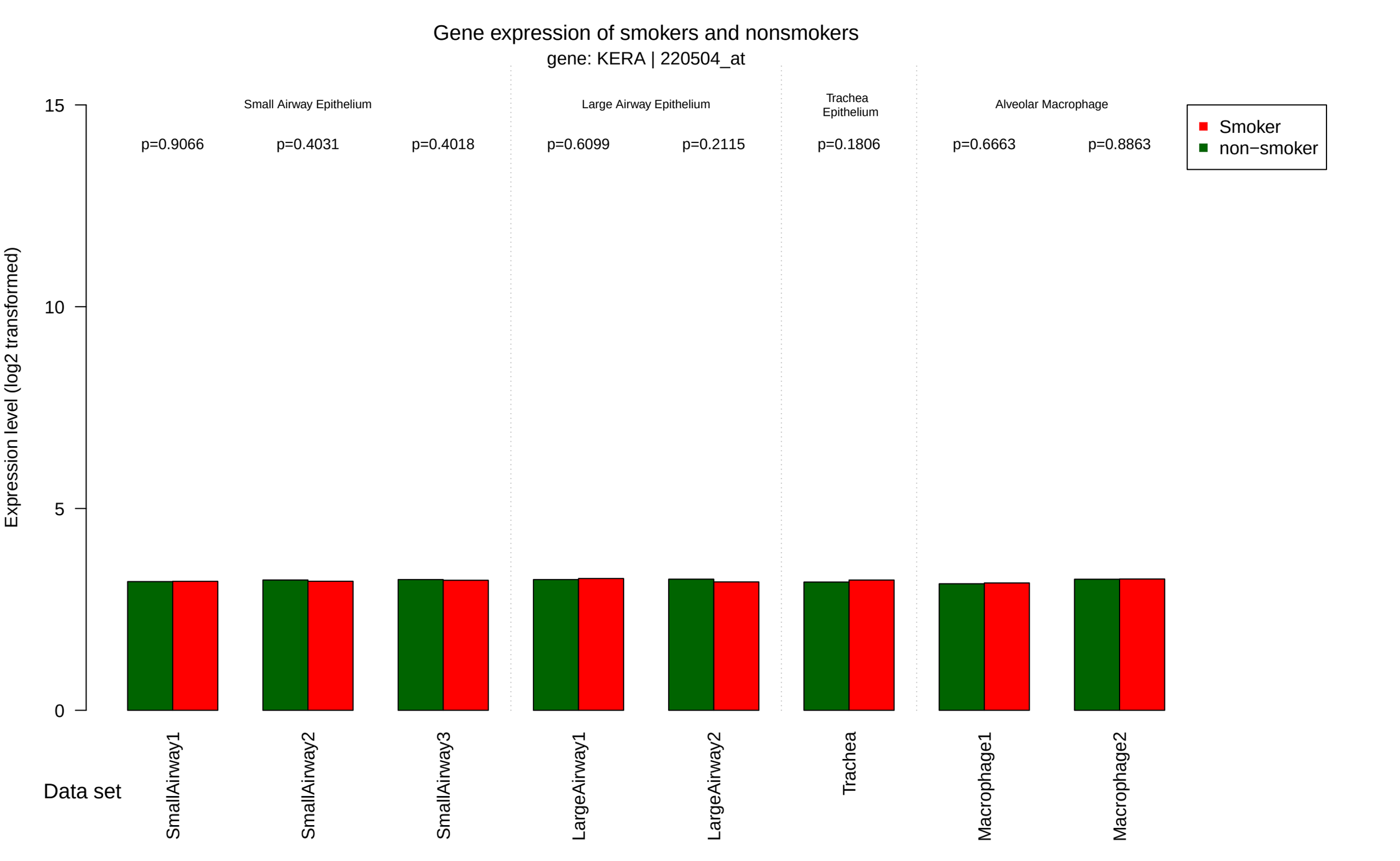
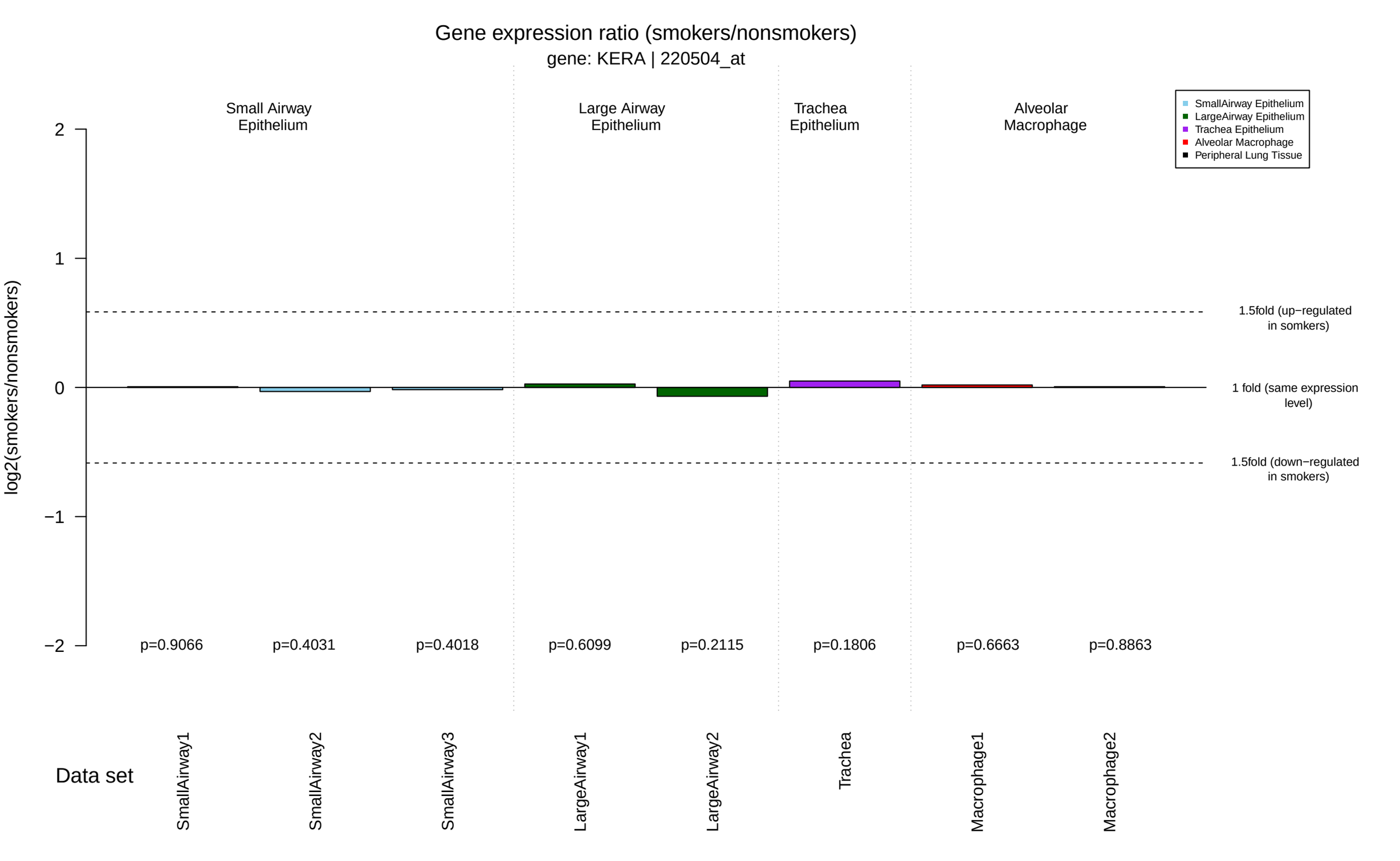
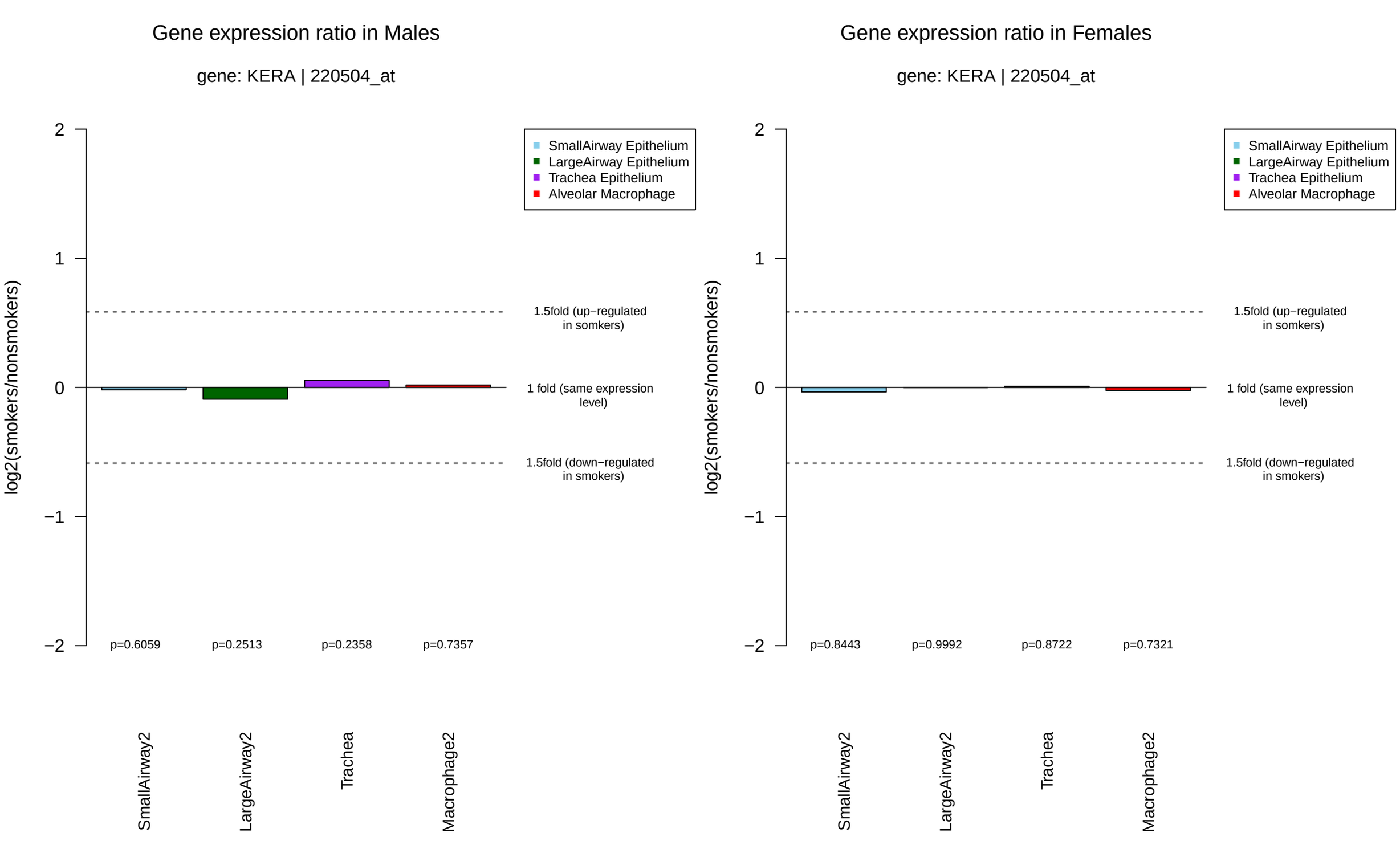
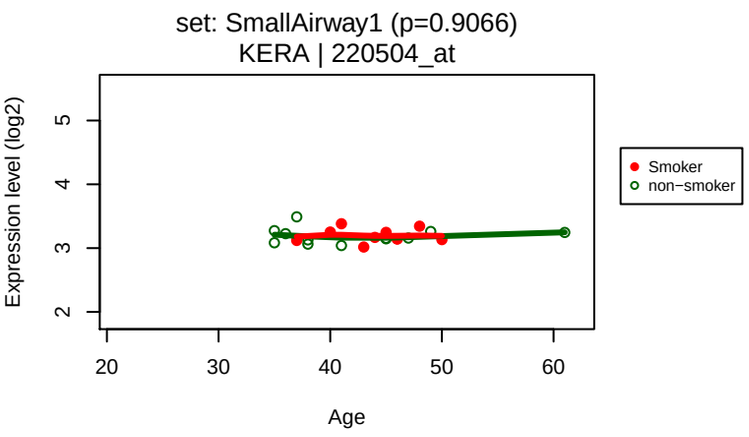
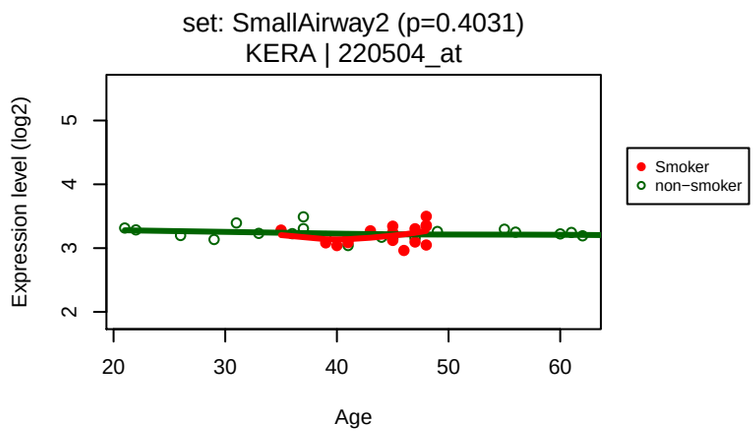
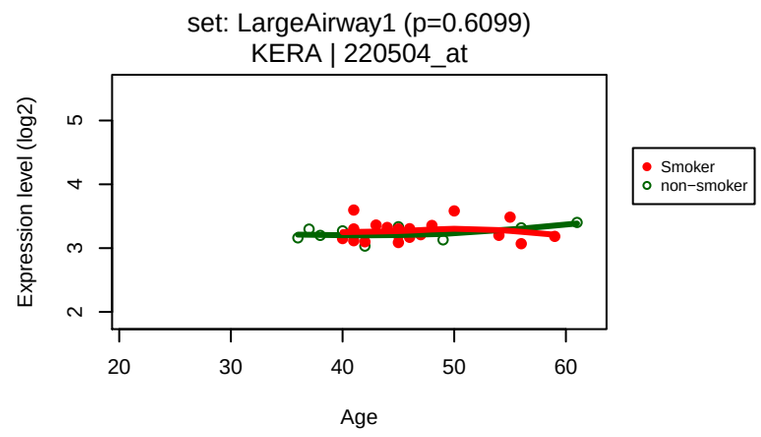

Gene expression of smokers and nonsmokers
gene: KERA | 220504_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9066
p=0.4031
p=0.4018
p=0.6099
p=0.2115
p=0.1806
p=0.6663
p=0.8863
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

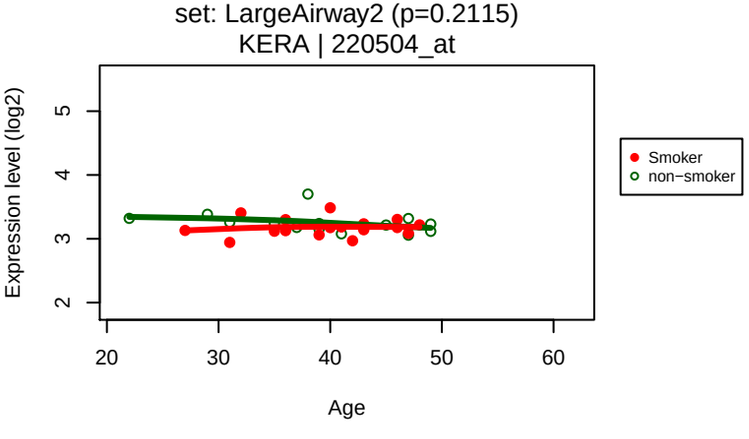

Gene expression ratio (smokers/nonsmokers)
gene: KERA | 220504_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9066
p=0.4031
p=0.4018
p=0.6099
p=0.2115
p=0.1806
p=0.6663
p=0.8863
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

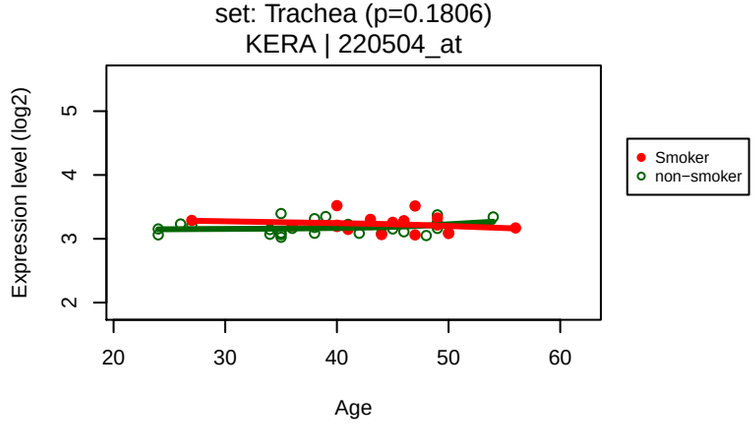

Gene expression ratio in Males
gene: KERA | 220504_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6059
p=0.2513
p=0.2358
p=0.7357
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KERA | 220504_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8443
p=0.9992
p=0.8722
p=0.7321
SmallAirway2
LargeAirway2
Trachea
Macrophage2

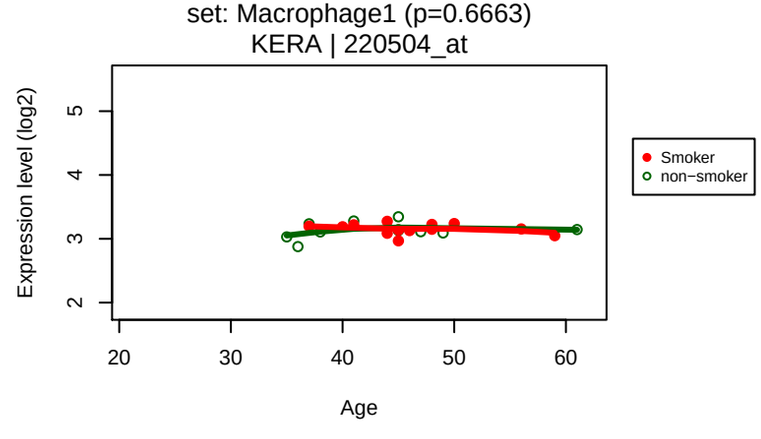

set: SmallAirway1 (p=0.9066)
KERA | 220504_at
Expression level (log2)
Age
Smoker
non−smoker

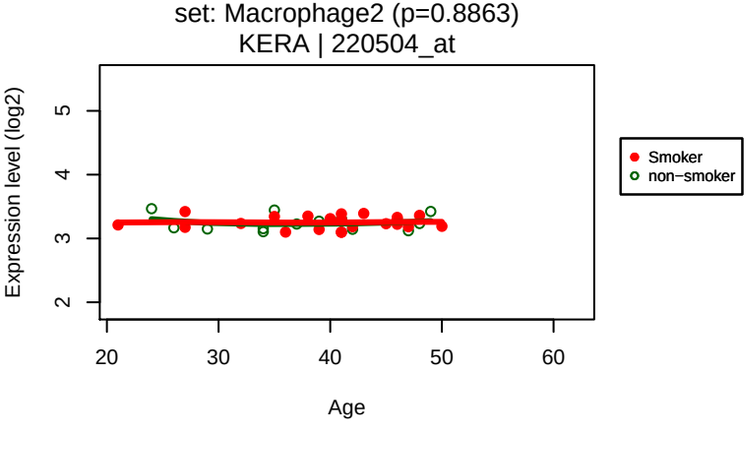

set: SmallAirway2 (p=0.4031)
KERA | 220504_at
Expression level (log2)
Age
Smoker
non−smoker

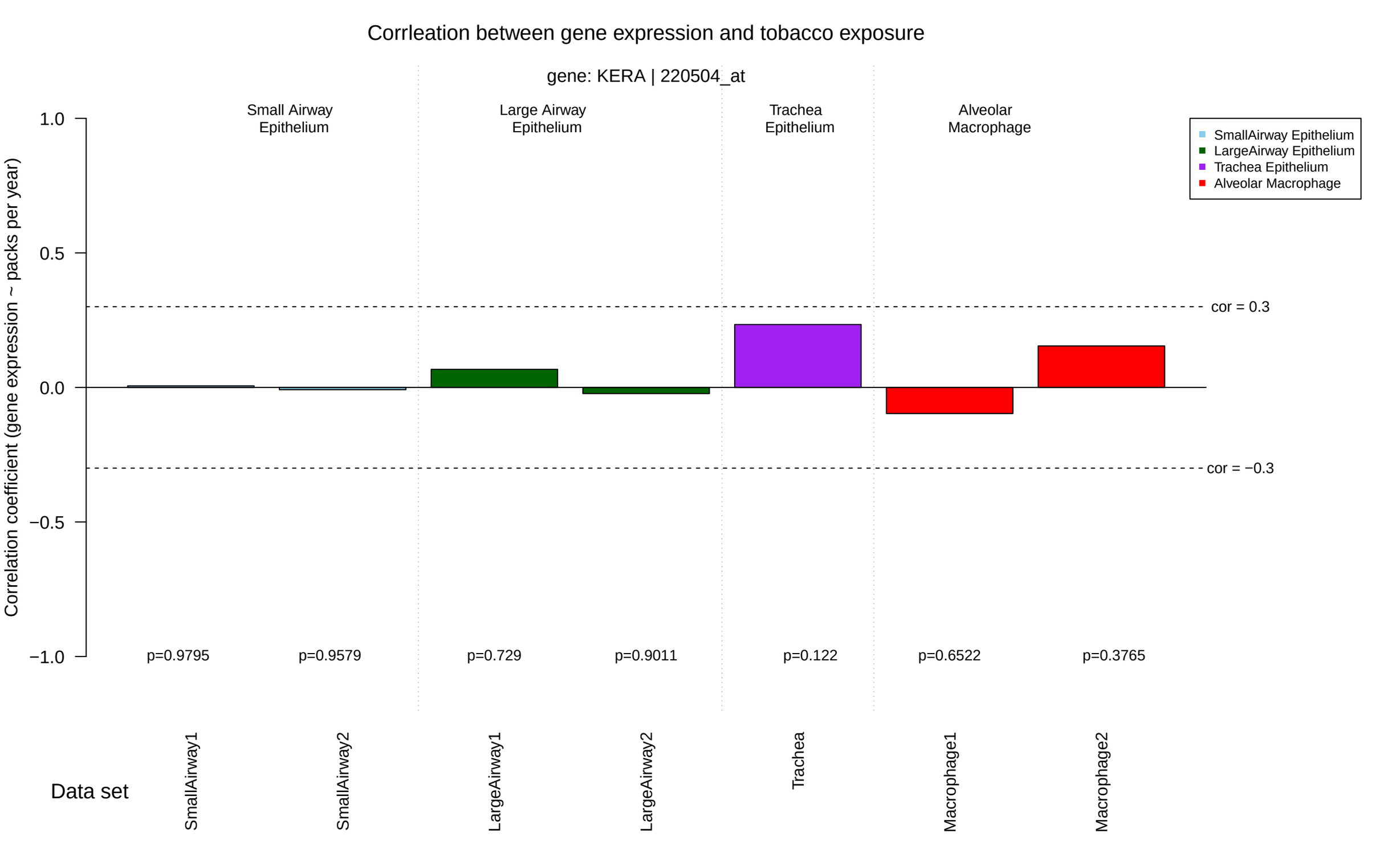

set: LargeAirway1 (p=0.6099)
KERA | 220504_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2115)
KERA | 220504_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1806)
KERA | 220504_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6663)
KERA | 220504_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8863)
KERA | 220504_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KERA | 220504_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9795
p=0.9579
p=0.729
p=0.9011
p=0.122
p=0.6522
p=0.3765
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2